



CATERPILLAR DEALER BEST PRACTICE

Truck 793D Hinge installation on the hydraulic tank cover

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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February 2021
1.10-210217-1

1.0 Introduction

We have in the routine of preventive maintenance to remove the cover of the hydraulic tank, to make cleaning inside the hydraulic tank; we evaluate and observe a risk in this activity. We then decided to develop a best practices, installing a hinge on the hydraulic tank cover has an ergonomic criterion and safety action.



View of the open tank, without removing the cover. View of the installation location of the hinges.

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2.0 Best practice description.

The purpose of installing a hinge on the cover of the hydraulic tank; It has the objective of reducing the risk, during the removal of the cover, to clean the internal part of the hydraulic tank, due to the weight (42kg), with difficult lifting point, time gain and simplifying the activity, being necessary only to be only one person is needed to do the job.



View of the open tank, with the aid of hinges and facility to open and close the hydraulic tank.

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3.0 Implementation steps.

This step is very simple; With the hydraulic tank empty, the cover in place, with the specific torque applied to the screws 30 lb / ft; weld two hinges (328-6156) one at each end of the cover so that the space for the cover fixing screws is free to be installed and removed..



Hinges installation location.

4.0 Benefits.

- Reduces the risk of injury when removing the cover from the hydraulic tank.
- Time savings during the cleaning activity of the inside of the hydraulic tank.
- Reduces the number of people to do the internal cleaning of the hydraulic tank.

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5.0 Resources required.

We can use the PM 4000 opportunity window where the hydraulic system oil is drained or repairs to the system, replacing the brake cooling pump, cleaning the tank, to apply the improvement.

One hour of the welder's labor, two hour of the mechanic and Parts total cost (~\$200.00).

6.0 Supporting attachments / References.

It is essential that all applicable safety measures and procedures are carried out. For more information refer to the operation, maintenance, service manual and general welding procedures as well as those measures outlined by the executing parties, mine regulations and associated agencies.

REHS1841 – General Welding Procedures: Welding on Fuel Tanks and Hydraulic Tanks

7.0 Related best practices.

Installed in a large part of the iron mine's 793D truck fleet, with excellent acceptance by the preventive maintenance team.

8.0 Acknowledgments.

Thanks to the engineering team and customer, for accepting the Project.

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